

CHANGBAI, China

WMO: 543860

Lat: 41.4249N

Lon: 128.1990E

Elev: 775

StdP: 92.35

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.8	-25.9	-32.1	0.2	-27.2	-30.1	0.3	-25.1	10.5	-14.3	8.8	-15.1	2.0	40	0.544

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	29.1	19.4	27.6	18.9	26.1	18.3	21.7	26.5	20.8	25.0	20.0	23.9	2.6	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.3	16.5	24.0	19.5	15.6	23.1	18.7	14.9	22.2	67.7	26.6	64.2	25.0	61.2	24.0	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min		Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.8	7.2	6.0		-30.7	31.7	2.3	1.6	-32.3	32.9	-33.7	33.8	-35.0	34.7	-36.7	35.9
			DB	-30.8	22.6	2.3	1.0	-32.5	23.4	-33.8	24.0	-35.1	24.5	-36.8	25.3
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.6	-16.3	-11.9	-3.7	5.2	11.8	16.9	20.6	19.7	13.4	5.8	-4.7	-14.0
	DBStd	13.48	4.45	5.16	5.25	4.62	3.65	2.79	2.40	2.84	3.35	4.21	5.76	5.00
	HDD10.0	3369	815	614	426	155	24	1	0	0	9	141	440	744
	HDD18.3	5524	1073	848	684	396	204	59	8	18	151	390	690	1003
	CDD10.0	1041	0	0	0	9	79	206	327	300	110	9	0	0
	CDD18.3	156	0	0	0	0	1	14	77	61	2	0	0	0
	CDH23.3	1178	0	0	0	4	71	205	515	358	24	0	0	0
	CDH26.7	246	0	0	0	0	12	36	123	74	1	0	0	0
Wind	WSAvg	2.2	2.0	2.2	2.9	2.9	2.8	2.2	1.9	1.9	2.0	2.2	2.0	1.9
Precipitation	PrecAvg	648	7	11	15	33	69	100	164	118	61	29	26	14
	PrecMax	852	21	39	31	68	153	176	328	216	143	55	70	48
	PrecMin	426	1	2	3	10	25	26	47	28	7	2	4	2
	PrecStd	105	5	8	8	15	28	40	63	47	37	16	15	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.3	4.9	14.1	23.8	28.3	29.6	31.8	30.8	25.7	20.9	11.8	0.1
		MCWB	-3.6	0.6	5.4	11.7	13.7	16.8	21.0	21.4	16.9	11.1	5.6	-2.3
	2%	DB	-4.6	1.9	9.9	20.1	25.3	27.6	29.5	28.5	23.6	18.1	8.5	-2.0
		MCWB	-6.6	-1.5	3.2	9.4	12.9	16.3	20.4	20.5	15.8	9.9	3.6	-3.9
	5%	DB	-6.8	-0.3	7.1	17.2	23.0	25.7	27.9	26.9	22.0	15.8	5.9	-4.0
		MCWB	-8.4	-3.0	1.7	8.0	11.9	16.0	20.0	20.0	14.8	8.6	2.3	-5.7
	10%	DB	-8.9	-2.5	4.6	14.4	20.6	23.8	26.2	25.3	20.3	13.5	3.4	-6.0
		MCWB	-10.2	-4.8	0.4	6.5	11.2	15.5	19.4	19.1	13.9	7.4	0.8	-7.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.7	1.1	6.1	12.6	16.0	19.1	23.0	22.7	19.2	12.3	6.6	-1.5
		MCDB	-1.8	3.6	12.6	21.7	24.7	25.2	29.1	27.8	22.8	18.4	10.3	-0.5
	2%	WB	-6.4	-1.2	3.7	10.1	14.4	18.1	22.1	21.7	17.6	10.7	4.3	-3.9
		MCDB	-4.8	1.6	9.0	18.4	22.0	24.0	27.3	26.2	21.6	16.3	7.5	-2.2
	5%	WB	-8.4	-2.9	2.1	8.5	13.1	17.2	21.2	20.9	16.1	9.4	2.7	-5.6
		MCDB	-6.9	-0.4	6.5	15.9	20.5	22.9	25.5	25.1	20.1	14.7	5.6	-4.0
	10%	WB	-10.2	-4.8	0.5	7.0	11.9	16.4	20.3	20.2	15.0	8.0	0.8	-7.5
		MCDB	-8.8	-2.4	4.2	13.7	19.1	21.9	24.3	24.0	18.7	12.6	3.1	-6.0
Mean Daily Temperature Range	5% DB	MDBR	11.9	13.2	12.2	13.2	13.8	12.3	10.2	10.3	13.3	13.5	11.1	11.1
		MCDBR	12.4	13.7	15.2	18.9	19.3	16.3	13.5	12.4	15.3	17.1	13.9	11.2
		MCWBR	10.6	10.7	9.4	9.7	8.4	6.6	5.4	5.1	7.2	9.3	9.2	9.5
	5% WB	MCDBR	12.0	12.7	13.9	17.4	16.0	12.8	10.5	10.1	11.7	15.2	12.5	10.9
		MCWBR	10.4	10.1	9.1	9.4	7.5	5.8	4.7	4.5	6.0	8.6	8.6	9.3
Clear-Sky Solar Irradiance	taub	0.323	0.384	0.460	0.532	0.567	0.584	0.523	0.474	0.421	0.421	0.369	0.329	
	taud	2.089	1.944	1.816	1.742	1.725	1.754	1.989	2.117	2.183	2.051	2.084	2.096	
	Ebn at Noon	795	791	771	746	733	721	764	791	808	744	729	748	
	Edh at Noon	108	148	190	220	229	223	175	149	130	132	109	100	
All-Sky Solar Radiation	RadAvg	2.48	3.46	4.48	4.86	5.23	5.02	4.68	4.58	4.26	3.33	2.27	2.03	
	RadStd	0.14	0.16	0.26	0.30	0.47	0.48	0.49	0.40	0.40	0.29	0.23	0.10	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.35	N/A	-1.13	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page